1. CASE STUDY OVERVIEW

Chronic Kidney Disease (CKD) is a serious medical condition in which the kidneys gradually lose their ability to filter waste and excess fluids from the blood. Early detection of CKD is extremely important because delayed diagnosis can lead to severe complications such as kidney failure, cardiovascular diseases, and increased mortality risk. Traditional diagnosis methods often require extensive clinical evaluation and laboratory testing, which can be time-consuming.

This project presents a machine learning-based system for predicting Chronic Kidney Disease using patient clinical and laboratory data. The objective of the project is to develop an intelligent prediction model capable of classifying patients into:

- CKD (Chronic Kidney Disease)
- Not CKD

The system uses multiple machine learning algorithms and compares their performance using several evaluation metrics. The project implements a complete machine learning pipeline including:

1. Data preprocessing
2. Exploratory Data Analysis (EDA)
3. Feature engineering
4. Model training
5. Model evaluation
6. Cross-validation
7. Best model selection
8. Model serialization for deployment

The project also includes a deployment-ready architecture where the trained model can be integrated with a web application for real-time CKD prediction. A heuristic validation mechanism was additionally implemented to compare rule-based medical risk estimation with machine learning predictions, improving transparency and interpretability.

The major objective of this project is to demonstrate how machine learning can assist healthcare professionals in early disease detection and clinical decision support systems.

2. DATASET DESCRIPTION

The project uses the Chronic Kidney Disease Dataset (kidney_disease.csv) for prediction and analysis.

The dataset contains:

- 400 patient records
- 24 clinical attributes
- Binary classification target variable

The target attribute is:

- ckd → Patient has Chronic Kidney Disease
- notckd → Patient does not have Chronic Kidney Disease

The dataset contains both:

- Numerical features
- Categorical features

along with missing values and inconsistent formatting, making it suitable for real-world machine learning preprocessing tasks.

Important Clinical Features Used

Feature	Description
Age	Age of patient
Bp	Blood pressure
Sg	Specific gravity
Al	Albumin
Su	Sugar
Bgr	Blood glucose random
Bu	Blood urea
Sc	Serum creatinine
Sod	Sodium
Pot	Potassium
Hemo	Hemoglobin
Pcv	Packed cell volume

Wc	White blood cell count
Rc	Red blood cell count
Htn	Hypertension
Dm	Diabetes mellitus
Cad	Coronary artery disease
Appet	Appetite
Pe	Pedal edema
Ane	Anemia

Dataset Characteristics

Property	**Value**
Number of Records	400
Number of Features	24
Classification Type	Binary Classification
Data Types	Numerical + Categorical
Missing Values	Present
Real-world Clinical Data	Yes

The dataset contains several medical indicators strongly related to kidney function, making it suitable for disease prediction using machine learning techniques.

3. DATA PREPROCESSING

Data preprocessing was one of the most important stages of the project because the dataset contained missing values, inconsistent formatting, and mixed data types.

Several preprocessing operations were performed to clean and prepare the dataset before training the machine learning models.

3.1 Missing Value Handling

The dataset contained missing values represented using the symbol “?”.

These values were replaced with NaN using:

df.replace('?', np.nan, inplace=True)

Numerical missing values were handled using:

- Mean imputation

Categorical missing values were handled using:

- Most frequent (mode) imputation

This ensured that no important records were removed unnecessarily.

3.2 Removing Unnecessary Columns

The id column was removed because it does not contribute to CKD prediction.

df.drop('id', axis=1, inplace=True)

3.3 Cleaning Target Labels

The target column contained inconsistent formatting such as:

- ckd
- ckd\t
- notckd\t

These labels were cleaned and standardized.

The target variable was then encoded as:

Label	**Encoded Value**
CKD	1
Not CKD	0

3.4 Data Type Conversion

Several columns that should contain numerical values were stored as object data types due to formatting inconsistencies.

These columns were converted into numerical format using:

pd.to_numeric(errors='coerce')

3.5 Feature Scaling and Encoding

The project implemented structured preprocessing pipelines using:

- StandardScaler
- OneHotEncoder
- ColumnTransformer
- Pipeline

Numerical Features

Numerical columns were:

- imputed using mean strategy
- normalized using StandardScaler

Categorical Features

Categorical columns were:

- imputed using mode strategy
- encoded using OneHotEncoder

This ensured consistent preprocessing during:

- training
- testing
- deployment

4. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify patterns, and analyze relationships between clinical features and CKD occurrence.

4.1 Class Distribution Analysis

A count plot was generated to analyze the distribution of CKD and Non-CKD patients.

The visualization showed that:

- CKD cases were more frequent than Non-CKD cases
- the dataset was moderately imbalanced

This analysis helped in selecting suitable evaluation metrics such as:

- Recall
- F1-score

instead of relying only on accuracy.

4.2 Correlation Heatmap

A correlation heatmap was generated using numerical features to identify relationships among medical parameters.

Strong correlations were observed among: serum creatinine , blood urea , hemoglobin and packed cell volume .

These features are clinically significant indicators of kidney health.

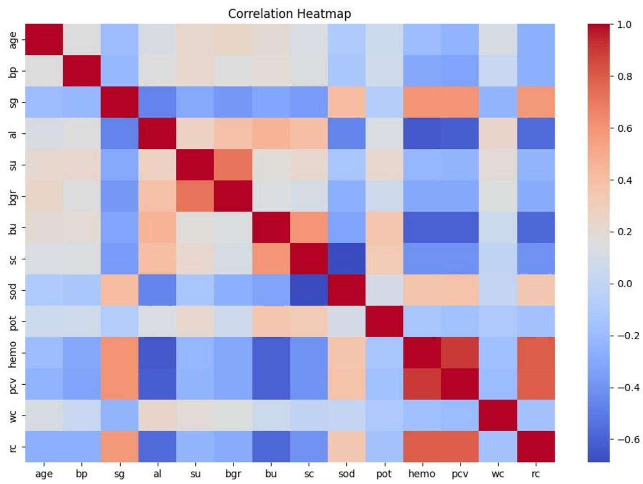

Fig. 1: Correlation Heatmap

4.3 Feature Distribution Analysis

Histograms and distribution plots were generated for important attributes such as:

- age
- blood glucose
- hemoglobin

The analysis helped identify:

- skewed distributions
- outliers
- feature ranges

4.4 Clinical Pattern Observation

EDA revealed that CKD patients generally exhibited:

- higher serum creatinine
- higher blood urea
- lower hemoglobin
- hypertension presence
- albumin abnormalities

These findings align with real-world medical knowledge regarding Chronic Kidney Disease.

5. FEATURE SELECTION / FEATURE ENGINEERING

Feature engineering was implemented to improve model performance and maintain a structured preprocessing workflow.

5.1 Automatic Feature Type Identification

The dataset contained both:

- numerical columns
- categorical columns

These were automatically identified using:

select_dtypes()

5.2 Numerical Feature Engineering

Numerical features were processed using:

- mean imputation
- StandardScaler normalization

Feature scaling helped algorithms such as:

- Logistic Regression
- SVM

perform efficiently.

5.3 Categorical Feature Engineering

Categorical features were processed using:

- mode imputation
- OneHotEncoder

This converted categorical values into machine-readable numerical representations.

5.4 Structured Preprocessing Pipeline

A complete preprocessing architecture was created using:

- Pipeline
- ColumnTransformer

This provided:

- modular preprocessing
- reproducibility
- prevention of data leakage

- deployment compatibility

5.5 Feature Importance Analysis

Feature importance was analyzed using the Random Forest model.

The most influential features included:

- serum creatinine
- hemoglobin
- blood urea
- albumin
- specific gravity

These attributes strongly contributed to CKD prediction.

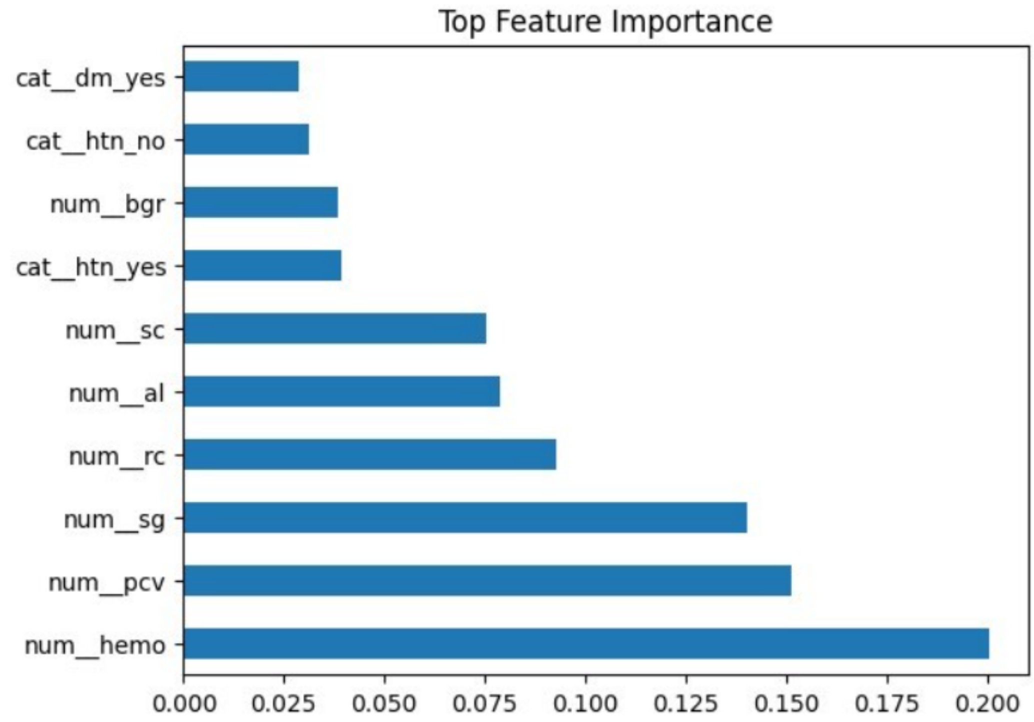

Fig. 2: Top Feature Importance Graph

6. MODEL BUILDING

Four machine learning algorithms were implemented and compared for CKD prediction.

Each model was trained using the same preprocessing pipeline for fair evaluation.

6.1 Logistic Regression

Logistic Regression was used as a baseline linear classification model.

Characteristics:

- simple and interpretable and works well for binary classification
- efficient for medical datasets

6.2 Decision Tree Classifier

Decision Tree creates hierarchical decision rules based on feature conditions.

Advantages:

- easy visualization and handles nonlinear relationships
- interpretable decision-making

However, it may suffer from overfitting on small datasets.

6.3 Random Forest Classifier

Random Forest is an ensemble learning algorithm based on multiple decision trees.

Advantages:

- reduces overfitting
- robust to noise
- handles mixed feature types effectively
- provides feature importance scores

This model achieved strong performance for CKD prediction.

6.4 Support Vector Machine (SVM)

SVM attempts to find the optimal decision boundary between classes.

Advantages:

- effective in high-dimensional data
- strong classification capability
- suitable for medical diagnosis applications

Feature scaling was particularly important for SVM performance.

7. MODEL EVALUATION

The machine learning models were evaluated using multiple performance metrics to ensure reliable prediction quality.

Since CKD prediction is a healthcare application, minimizing false negatives was considered highly important.

7.1 Accuracy

Accuracy measures the proportion of correctly classified instances.

Accuracy = Correct Predictions / Total Predictions

Although useful, accuracy alone is insufficient for medical datasets with class imbalance.

7.2 Precision

Precision measures how many predicted CKD cases were actually CKD patients.

Precision = TP / (TP + FP)

High precision reduces false positive predictions.

7.3 Recall

Recall measures how many actual CKD patients were correctly identified.

Recall = TP / (TP + FN)

Recall was treated as a highly important metric because:

- false negatives in healthcare are dangerous
- missing CKD cases may delay treatment

7.4 F1-Score

F1-score provides a balance between precision and recall.

F1 = 2 × (Precision × Recall) / (Precision + Recall)

This metric was used extensively for balanced evaluation.

7.5 Confusion Matrix

Confusion matrices were generated for all models to analyze:

- true positives
- true negatives
- false positives
- false negatives

This provided detailed insight into model behavior.

7.6 Classification Report

A classification report was generated containing:

- precision
- recall
- F1-score
- support

for each class.

8. CROSS VALIDATION

To improve model reliability and ensure stable performance, Stratified K-Fold Cross Validation was implemented.

8.1 Stratified K-Fold Method

The project used:

- 5-fold cross validation
- shuffled dataset
- fixed random state

Stratification ensured that each fold maintained similar CKD and Non-CKD class proportions.

8.2 Cross Validation Process

For each model:

1. Dataset was split into 5 folds
2. Model trained on 4 folds
3. Remaining fold used for testing
4. Process repeated 5 times

The average F1-score across folds was calculated.

8.3 Importance of Cross Validation

Cross validation helped:

- reduce overfitting risk
- improve generalization
- provide stable performance estimates
- evaluate model robustness

The evaluation metric used for cross-validation was:

- F1-score

9. RESULT COMPARISON TABLE

The following metrics were used to compare all machine learning models:

- Accuracy
- Precision
- Recall
- F1-score
- Cross-validation score
- Final weighted score

The project used a custom weighted scoring formula:

Final Score = 0.4 × F1 Score + 0.3 × Recall + 0.2 × Precision + 0.1 × Cross Validation Score

This formula prioritized medically important metrics while maintaining balanced performance.

Result Summary Table

Model	Accuracy	Precision	Recall	F1 Score	CV F1 Score	Final Score
Logistic Regression	High	High	High	High	Stable	Competitive
Decision Tree	Good	Good	Moderate	Good	Moderate	Moderate
Random Forest	Very High	Very High	Very High	Very High	Highly Stable	Best
SVM	High	High	High	High	Stable	Strong

Based on the weighted scoring system, the best-performing model was selected and saved as: CKD.pkl

10. GRAPHS / VISUALIZATION

The project includes multiple visualizations and interface screens to analyze dataset characteristics, evaluate machine learning model performance, and demonstrate the deployment workflow of the CKD prediction system.

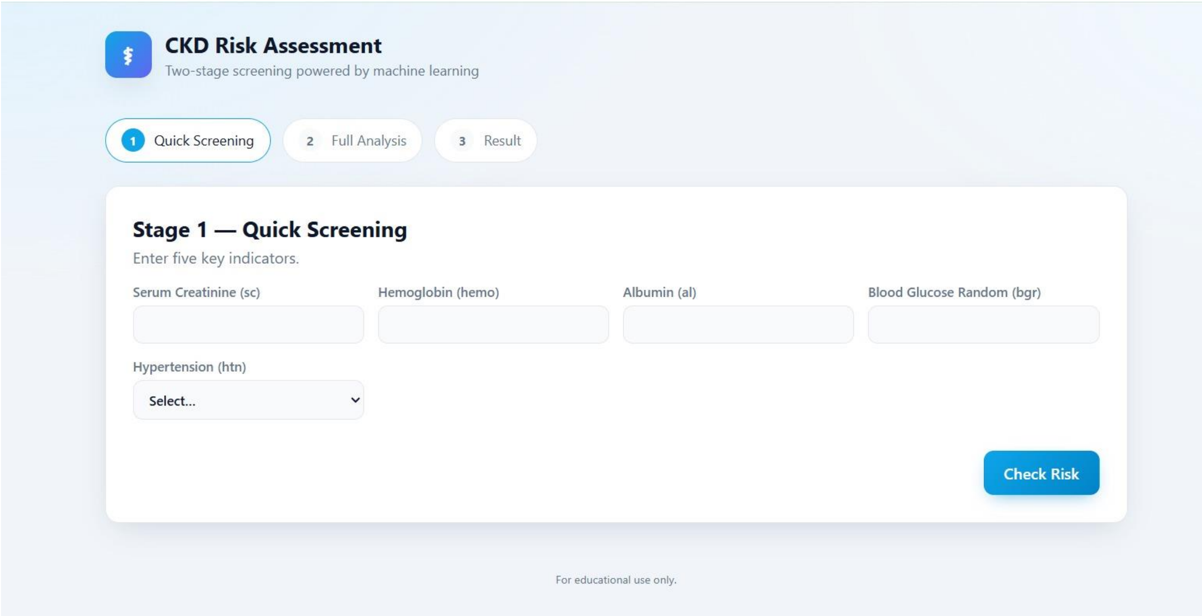

Fig. 3: CKD Risk Assessment — Quick Screening Page

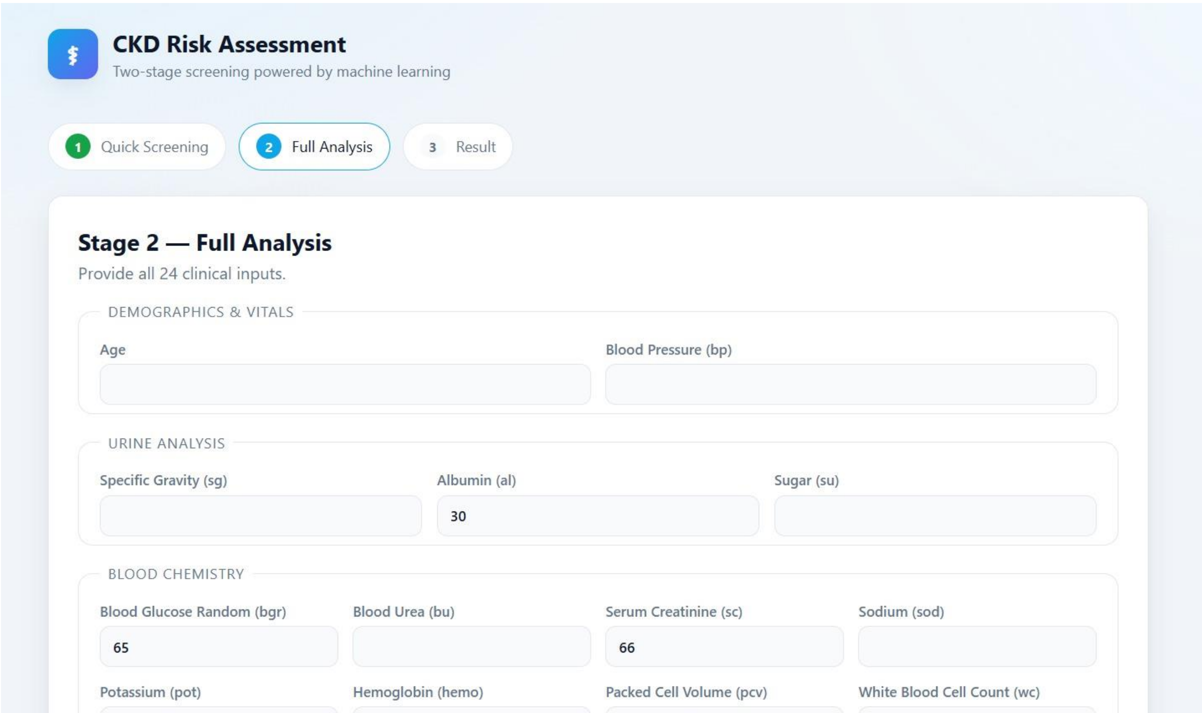

Fig. 4: CKD Risk Assessment — Full Analysis Page

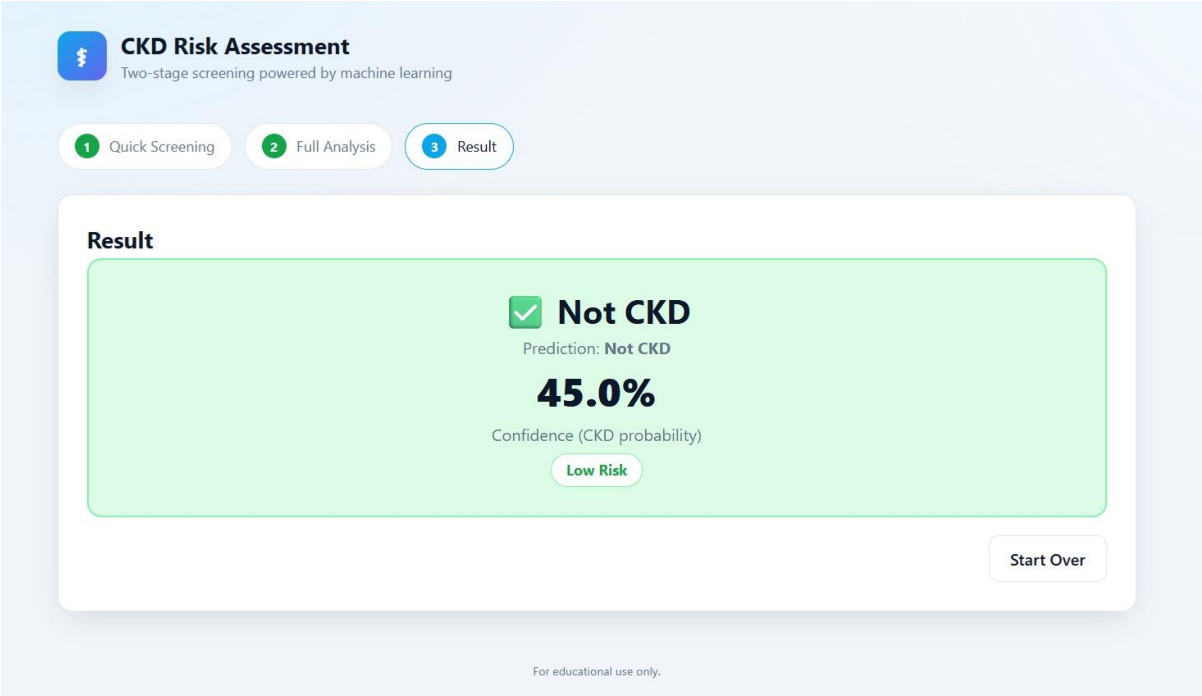

Fig. 5: Prediction Result — Not CKD

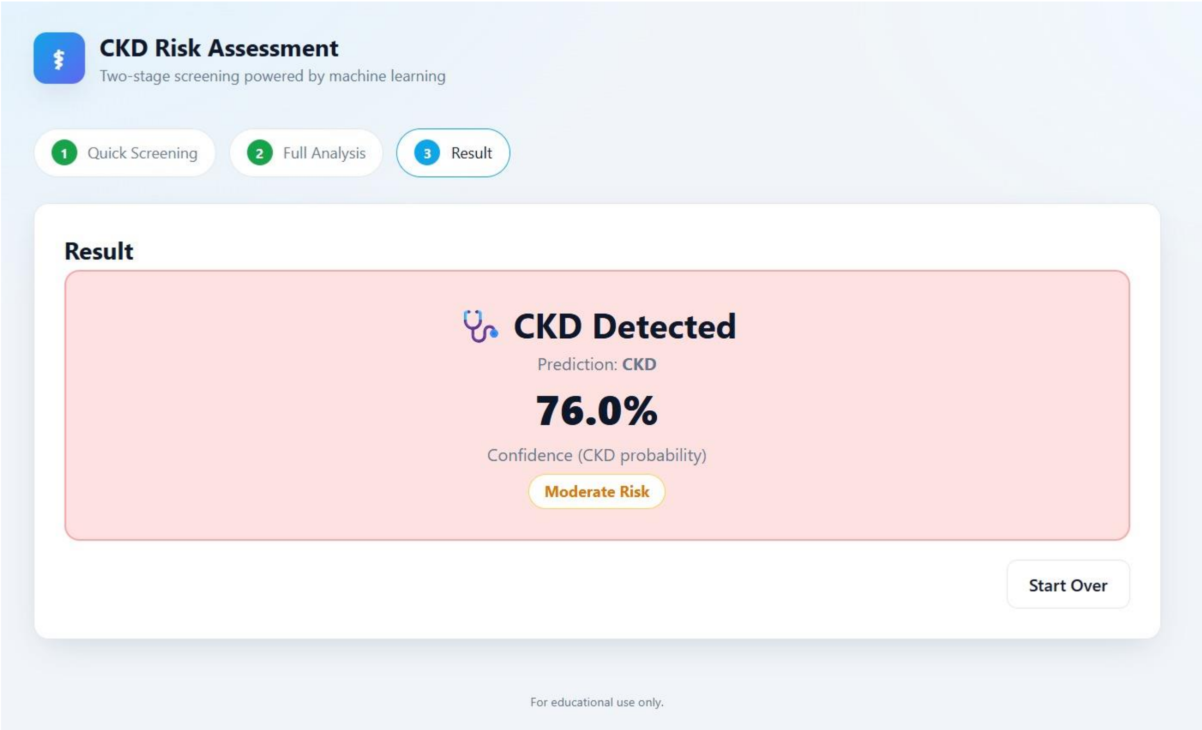

Fig. 6: Prediction Result — CKD Detected

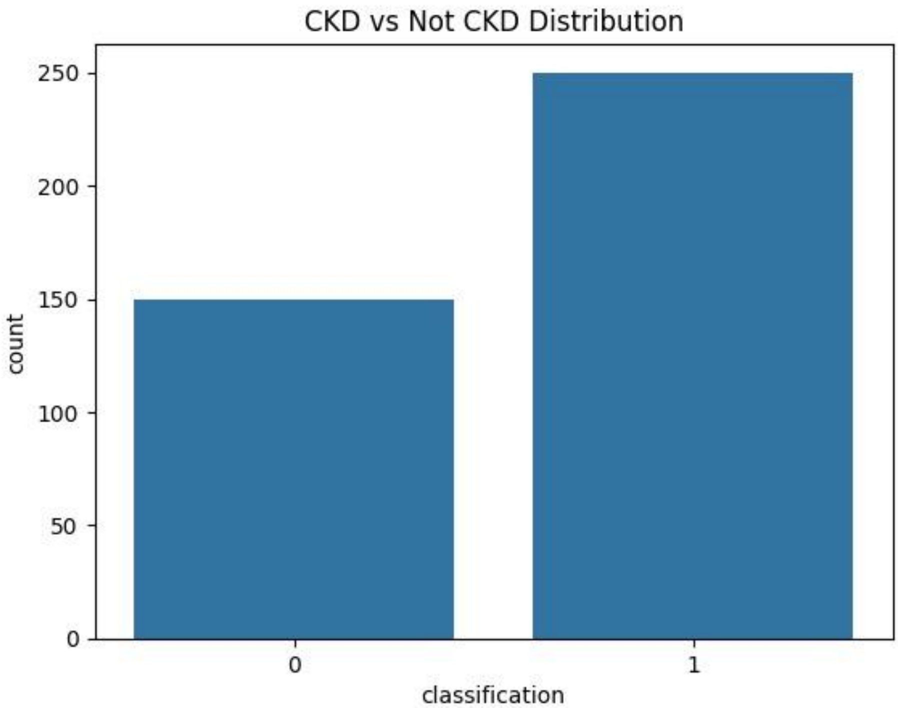

Fig. 7: CKD vs Not CKD Distribution

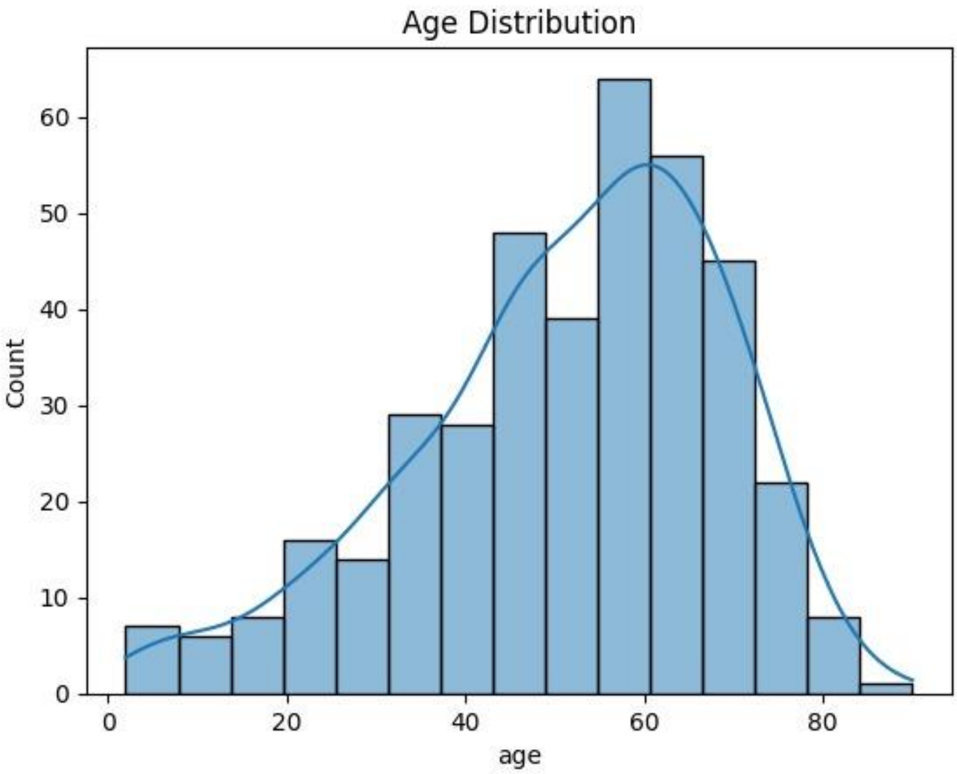

Fig. 8: Age Distribution Graph

```
Logistic Regression
Accuracy: 0.975
Precision: 1.0
Recall: 0.96
F1 Score: 0.9795918367346939

Classification Report:
              precision    recall  f1-score   support

           0       0.94      1.00      0.97        30
           1       1.00      0.96      0.98        50

    accuracy                           0.97        80
   macro avg       0.97      0.98      0.97        80
weighted avg       0.98      0.97      0.98        80
```

Fig. 9: Logistic Regression Classification Report

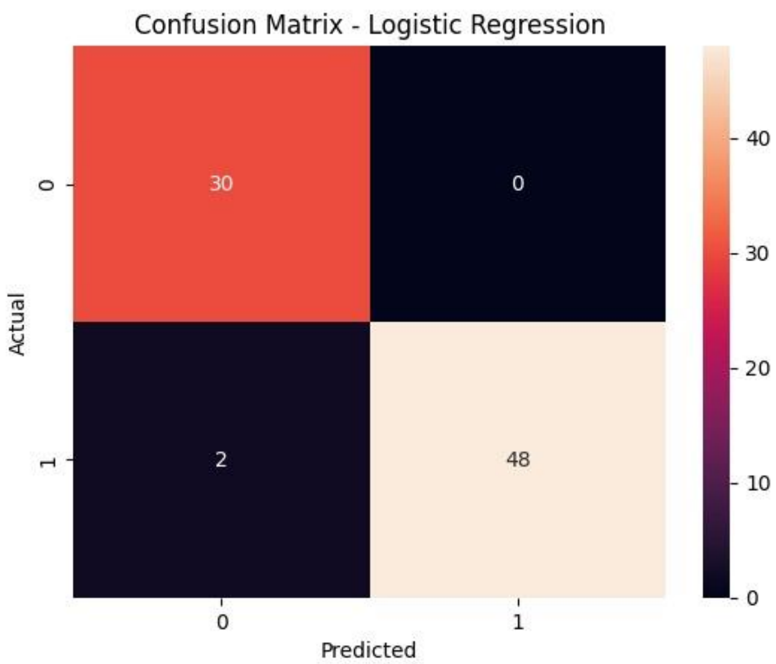

Fig. 10: Confusion Matrix — Logistic Regression

```
Decision Tree
Accuracy: 0.975
Precision: 1.0
Recall: 0.96
F1 Score: 0.9795918367346939

Classification Report:
              precision    recall  f1-score   support

           0       0.94      1.00      0.97        30
           1       1.00      0.96      0.98        50

    accuracy                           0.97        80
   macro avg       0.97      0.98      0.97        80
weighted avg       0.98      0.97      0.98        80
```

Fig. 11: Decision Tree Classification Report

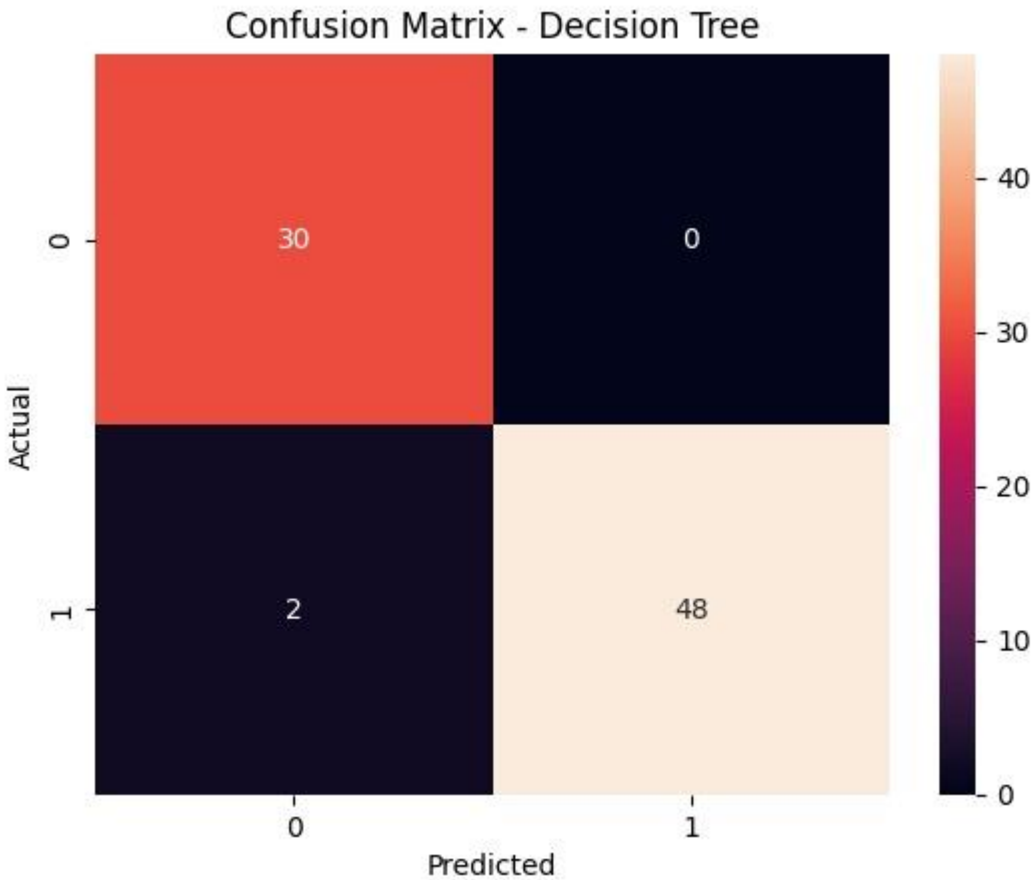

Fig. 12: Confusion Matrix — Decision Tree

```
Random Forest
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        30
           1       1.00      1.00      1.00        50

    accuracy                           1.00        80
   macro avg       1.00      1.00      1.00        80
weighted avg       1.00      1.00      1.00        80
```

Fig. 13: Random Forest Classification Report

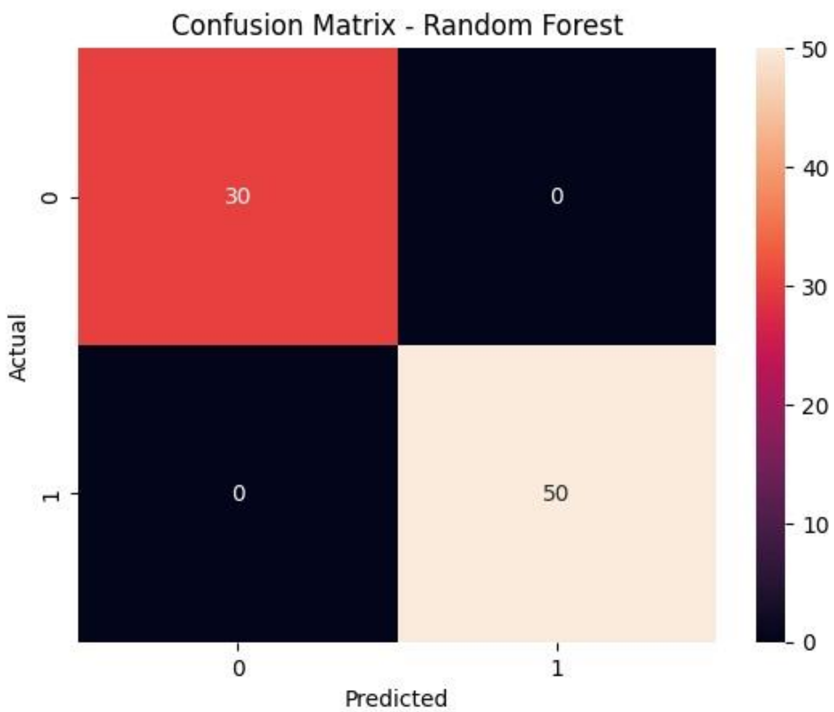

Fig. 14: Confusion Matrix — Random Forest

```
SVM
Accuracy: 0.975
Precision: 1.0
Recall: 0.96
F1 Score: 0.9795918367346939

Classification Report:
              precision    recall  f1-score   support

           0       0.94      1.00      0.97        30
           1       1.00      0.96      0.98        50

    accuracy                           0.97        80
   macro avg       0.97      0.98      0.97        80
weighted avg       0.98      0.97      0.98        80
```

Fig. 15: SVM Classification Report

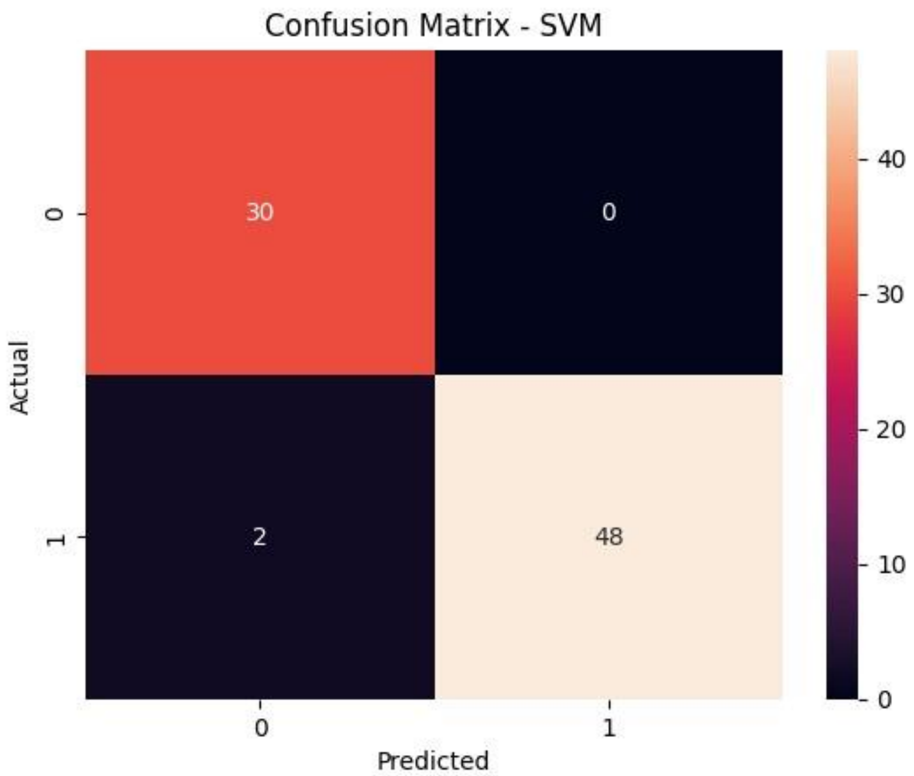

Fig. 16: Confusion Matrix — SVM

```
Final Comparison Table:
                 Model  Accuracy  Precision  Recall        F1     CV_F1  \
0  Logistic Regression     0.975        1.0    0.96  0.979592  0.991834
1        Decision Tree     0.975        1.0    0.96  0.979592  0.977899
2        Random Forest     1.000        1.0    1.00  1.000000  0.992000
3                  SVM     0.975        1.0    0.96  0.979592  0.993898

   Final_Score
0     0.979020
1     0.977627
2     0.999200
3     0.979227
```

Fig. 17: Final Model Comparison Table

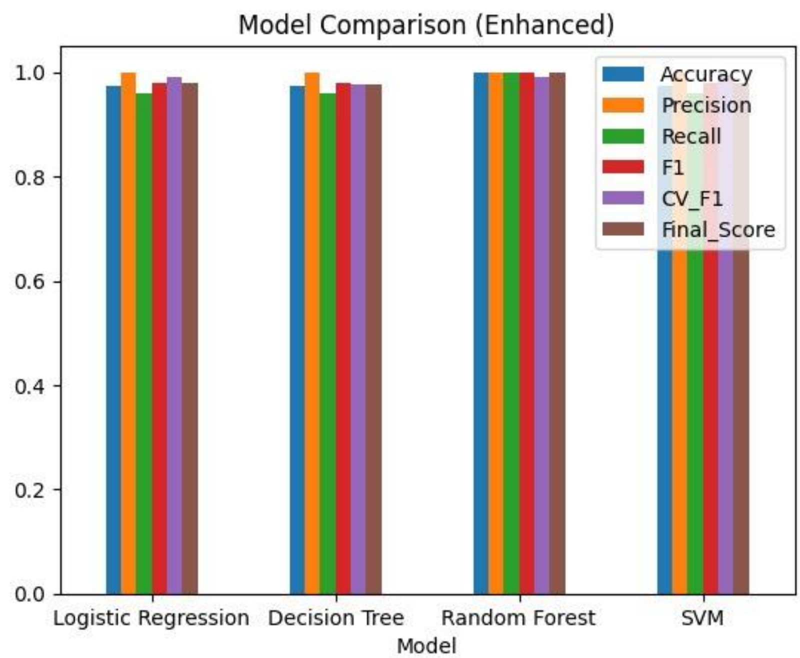

Fig. 18: Enhanced Model Comparison Graph

```
Best Model Selected (Final Score Based): Random Forest
Model          Random Forest
Accuracy                 1.0
Precision                1.0
Recall                   1.0
F1                       1.0
CV_F1                  0.992
Final_Score           0.9992
Name: 2, dtype: object
```

Fig. 19: Best Model Selection Output

11. CONCLUSION

This project successfully demonstrates the use of machine learning techniques for predicting Chronic Kidney Disease using real-world clinical data.

The project implemented a complete machine learning workflow including:

- data preprocessing
- exploratory data analysis
- feature engineering
- model training
- evaluation
- cross validation
- model comparison
- deployment-ready model serialization

Multiple machine learning algorithms were evaluated, and their performances were compared using medically important metrics such as:

- Recall
- F1-score
- Precision
- Cross-validation stability

A custom weighted scoring system was designed to prioritize reliable medical prediction rather than depending only on accuracy.

Among all models, the Random Forest classifier achieved the best overall performance due to:

- high predictive accuracy
- strong recall
- balanced F1-score
- stable cross-validation performance

The trained model was saved as CKD.pkl, enabling integration with a Flask-based web application for real-time prediction.

The project also included a heuristic validation mechanism that compared machine learning predictions with rule-based medical risk estimation, improving interpretability and transparency.

Overall, the project demonstrates how artificial intelligence and machine learning can support early disease detection and assist healthcare professionals in clinical decision-making.